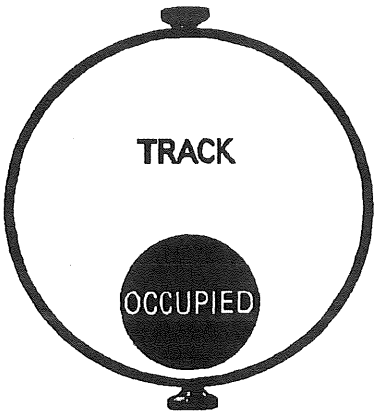
INTRODUCTION TO SIGNALLING

8

TRAIN DETECTION



A.J.R.Rowbotham

INTRODUCTION

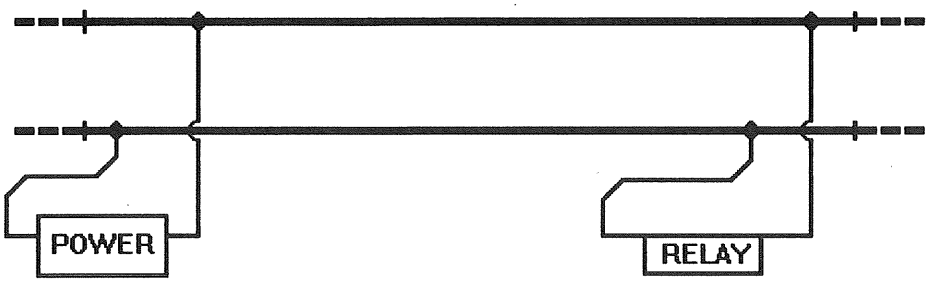
An essential requirement in signalling is the need to determine whether a defined section of track is clear of traffic, or has any vehicles (stationary or moving) on it. One method, introduced in the UK around the turn of the century, is the *track circuit*, using electrical circuits in the rails. A more recent method is the *axle counter,* which has the advantage over track circuits in certain circumstances.

THE BASIC TRACK CIRCUIT

The track circuit is an absolutely essential component of signalling and life would be somewhat difficult today without it. The track circuit is one means of detecting whether a section of track has any railway vehicles standing on it or not. Electrical circuits in the rails themselves are used for this task of detecting whether a section of track is *clear* or *occupied.* Officially the equipment is known as a *track circuit* but this is commonly shortened to just 'track'.

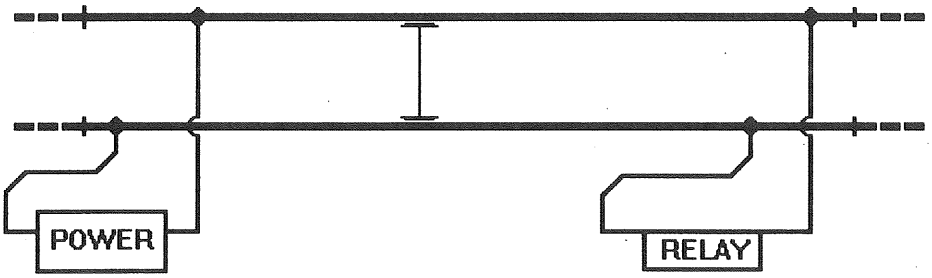
Below are shown two possible ways of organising a track circuit.

One way would be to connect a power supply across the rails at one end of the section, and connect a relay across the rails at the other end:



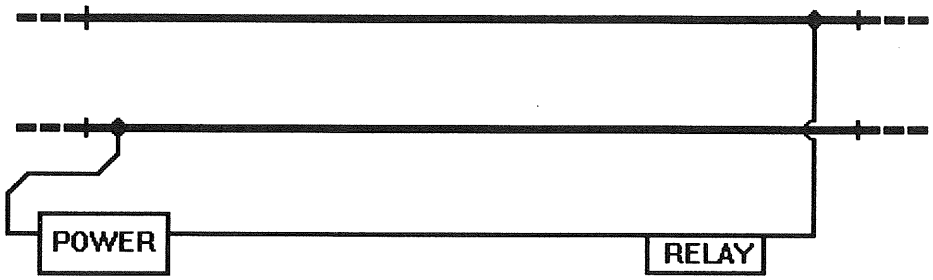
If the track circuit is clear, then the current will flow from the power source, along one rail, through the relay coil, along the other rail and back to the power source. Thus, if the track circuit is clear then the relay will be energised.

When a train occupies the track circuit:



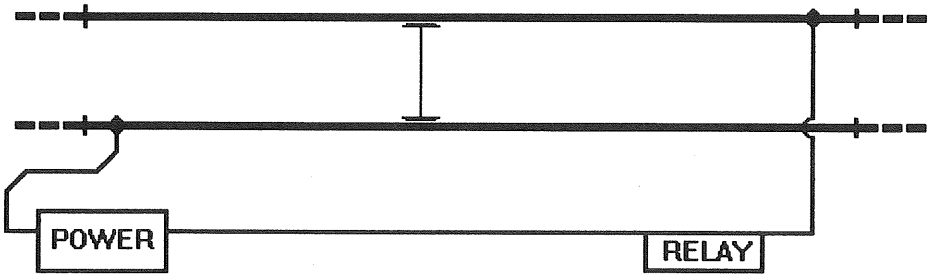
then the metal wheels will connect the two rails together. The current will then flow quite happily through the wheels and axles rather than having to push through the relay. (The resistance of the wheels/axles is very small in comparison with the resistance of the relay coil.) Thus if the track circuit is occupied, negligible current flows through the relay coil and the relay is de-energised.

An alternative method would be to put the power source and relay in series and connect them across the rails:



When the track is clear, the current will not be able to flow as there is not a complete path from the power source, through the circuit and back to the source. Thus if the track circuit is clear, the relay will be de-energised.

When the track circuit is occupied:



then the circuit is completed and the current will flow around the circuit through the relay coil. Thus, if the track circuit is occupied the relay will be energised.

Thus for the second suggested track circuit the relay works 'the other way up' from the first suggestion, i.e. in the first suggestion the relay is energised when the track is clear, in the second the relay is de-energised when the track is clear.

It is now necessary to go through a process of deciding which, if any, is the better choice. To do this one or two questions have to be asked:

Firstly - Does the track circuit have to behave in a fail-safe manner? It is essential to know without doubt if a section of track is occupied or not, so the answer is 'Yes'.

If the answer is 'yes', how can it be determined if the design *is* fail-safe or not?

Questions can be asked such as what happens if the power supply fails?

what happens if a wire becomes disconnected?

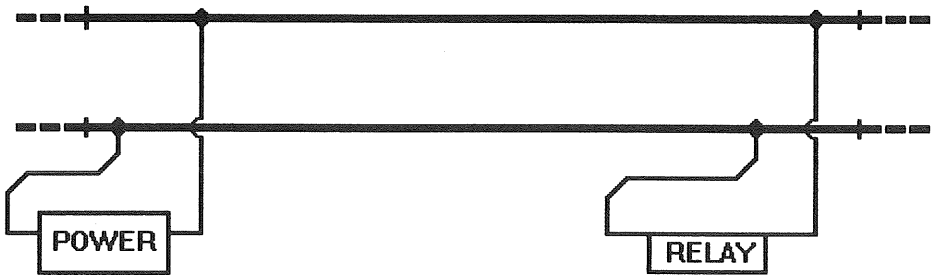
Considering the first suggested circuit, if the power fails or a wire breaks, the relay must be de-energised irrespective of the state of the track circuit. Under failure conditions the track circuit will then appear to be occupied.

Considering the second suggested circuit, again if the power fails or a wire breaks, the relay must be de-energised. This time however the system will infer that the track circuit is clear.

This would lead to a very dangerous situation - the track circuit has failed, but the system assumes the track is clear (even though it might be occupied). This is consequently quite unacceptable and is *not* fail-safe. This design of track circuit must never be used or even considered.

The first suggestion is fail-safe. If there is a failure the system will assume the track is occupied (even though it might be clear). This is fail-safe - the system must fail to the worst case - so this type of track circuit must always be used. One of the uses of track circuits is that a signal cannot show a proceed aspect unless all the track circuits along its route are proved to be clear. Thus, if a track circuit fails, all signals reading over that track will be held at danger. Trains may well be announced as being delayed 'because of a track failure'. This does not mean that the railway track has fallen to pieces, just that a track circuit has failed. The commuters may not be very happy, but it is for their own good.

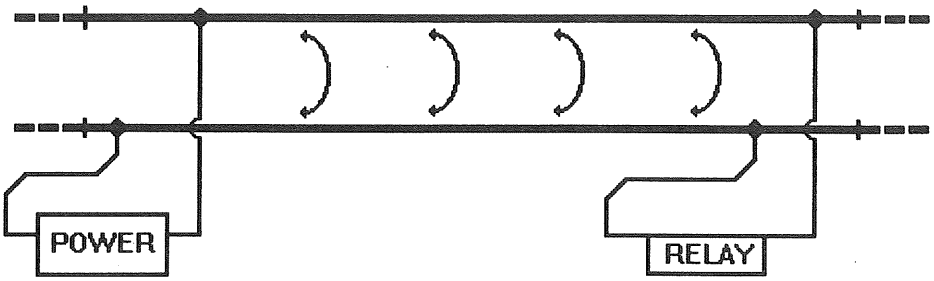
The basic safe design of the track circuit is consequently:



with the relay energised as long as the track circuit is clear and de-energised otherwise. Adjacent track circuits are separated from each other by *insulated joints* in the rails, thus stopping the adjacent tracks circuits from affecting each other. These joints are shown on the diagrams above by the symbol



The basic design of track circuit thus appears to be very simple. A power source is connected across the rails at one end of the section, and a relay at the other. Life is not that simple however. It has been assumed that the two rails are completely isolated from each other, but this is not the case. The rails are held in place by sleepers, which are in turn held in place by ballast. This combination is by no means a proper insulator, and current does in fact flow from one rail to the other through the ballast and sleepers.

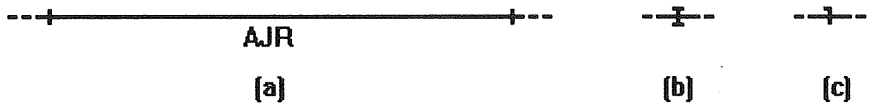


The amount of current flowing in this way varies with the conditions. If the ballast is contaminated with oil then a better path for this leakage current results. Similarly, more current flows if the ballast is wet after rain has fallen. The longer the track circuit the more the current from the power source will flow through the ballast rather than through the relay. A final problem is that the wheels and axles do have a small resistance rather than the ideal zero figure. The actual figures depends on how well the wheels are making contact with the rails, whether there are leaves on the line etc. Thus the effect of this leakage current varies all the time and this makes the design of track circuit components quite complex. This in turn puts restrictions on the application of the track circuit. Firstly, in order to limit the effects of the leakage current, a maximum length of track circuit is specified. Secondly, because the electrical parameters of the track circuit are critical, the lengths of the tail cables between the rails and the trackside location case housing the track circuit components are limited to specified measurements.

THE SCHEME PLAN

On each signalling scheme each track circuit must have a unique identity or name. Once upon a time each track circuit was given a number. Nowadays the standard is to use groups of letters for each. On a small scheme, pairs of letters are sufficient, e.g. SD or AH. On bigger schemes involving more than one interlocking, three letters are used, with the first letter designating the interlocking the track is involved with, e.g. ASD, AAH in interlocking 'A', BSD, BAH in interlocking 'B' etc.

On track plans the insulated joints separating one track circuit from the adjacent one is shown by a small bar across the track as shown in (a) below.



The identity of the track circuit ('AJR' in this case) is shown if possible below the line of track, and roughly in the middle of the two insulated joints. In some cases there may be something else on the track plan in the way, so it may be necessary to deviate from the rules. There must however be no doubt as to the identity of any section of track.

It is convenient to show on track plans the extent of the overlap at each signal. This is done by using the special symbol shown in (b) above for the appropriate overlap joint, rather than the conventional symbol. The different symbol does not infer that the joint is any different physically - it is used solely to give extra information on the track plan.

On occasions there will be the situation where there is a track circuit on one side of a joint and no track circuit on the other. In this situation the symbol shown in (c) above is used. The offset part at the top of the symbol points to the side where there is a track circuit, i.e. in the example there is a track circuit on the left of the joint, but no track circuit on the right hand side.

The end of the track where the power is applied is known as the *feed end*, while the relay end is, not surprisingly, known as the *relay end*. This information may be shown on the track plan by the letters 'F' and 'R' respectively:



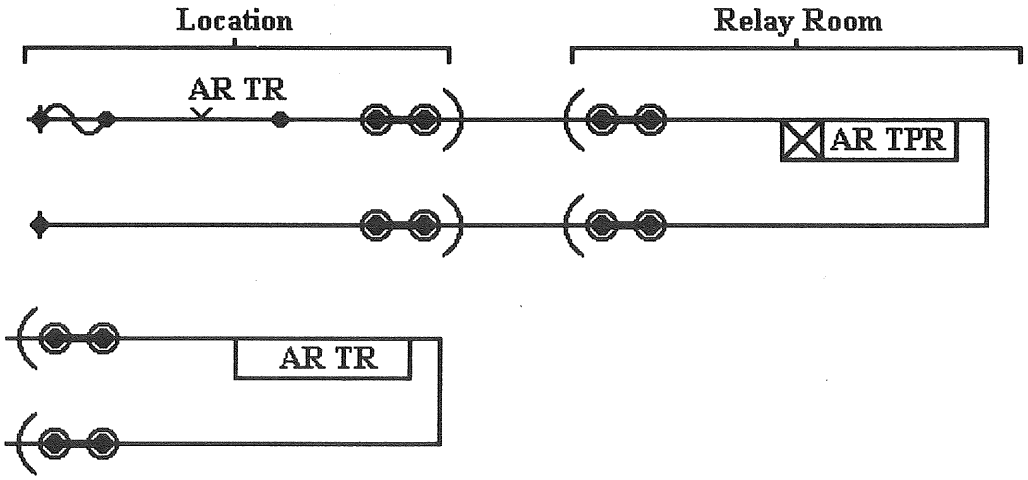
This shows that the feed end of track AJR is at the left hand track joint, both the relay end of AJR and the feed end of AJS are at the middle joint and the relay end of AJS is at the right hand joint.

USING THE TRACK CIRCUIT - THE PRACTICAL CIRCUIT

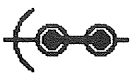
Consider the situation below, with a track circuit AR on a stretch of track, some distance from the relay room.



The relay will be housed in a Location close to the insulated joint (to keep the length of the track tail cable short), some way away from the relay room where the information about the state of the track is wanted. A circuit has to be organised between the Location and the Relay Room to repeat the information to where it is required. The power supply for this circuit will be in the Location, whilst the new repeater relay will be in the Relay Room. A two wire circuit is required so that the negative side of the relay can be brought back to the power supply in the location.



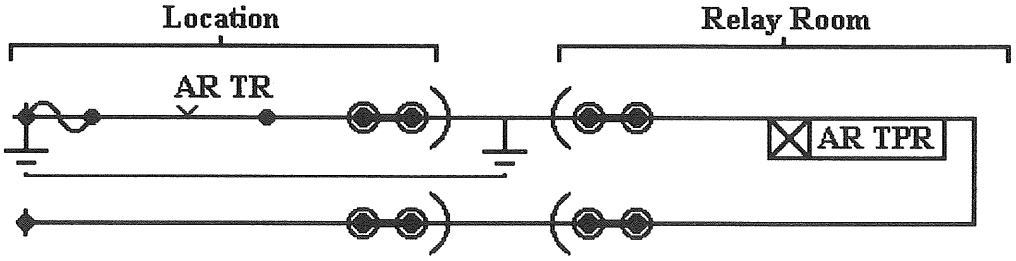
Conventionally the Location circuits would only show what is in the Location, and similarly, the Relay Room circuits only show what is in the Relay Room. This type of circuit, showing both 'ends' together is known as a *through circuit.* Location cases are normally wired in the factory, so all wiring for circuits entering or leaving the case is terminated on terminal blocks with sliding links, with the symbol:



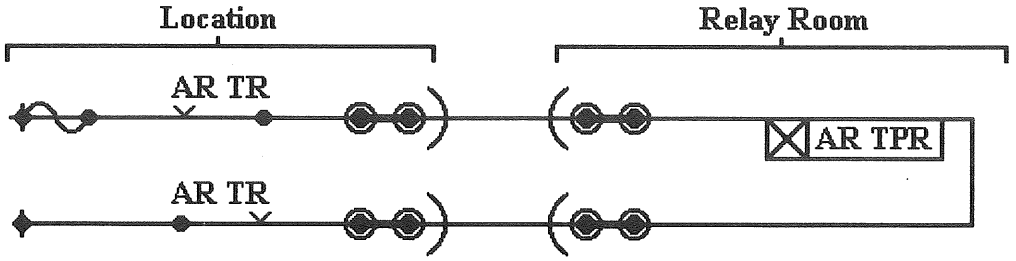
The arc symbol indicates that the circuit is entering or leaving a Location Case or other housing such as a Relay Room. The track relay is shown in the Location circuits, but the rail connections of the tail cables are not shown. The wires between the Location and the Relay Room terminal blocks would be part of one of the trackside multicore signalling cables.

It is common while a track is occupied for the Track Relay to pick up momentarily and then drop again, due to perhaps dirt on the rail, or the vehicle bouncing slightly. This problem is known as *track bobbing*. The bobbing of the Track Relay cannot be prevented, but to overcome the problem, a special type of relay is used for the repeater relay, the symbol for which has a St Andrew's Cross at the left hand end. This represents a slow pickup relay. That is the type where there is a delay before the contacts start moving after the coil is energised. If bobbing does occur, the current through the Track Relay contact current would have disappeared again before the contacts actually moved. The slow feature thus filters out the bobbing. It is a requirement that the slow pickup repeater must always be wired in before the track circuit is used in any signalling control. The contacts on the Track Relay must only be used to control this repeater. If there is a need to use the track circuit control in the Location case, then the repeater will also have to be in the Location case, and contacts of it will be used locally, and used to drive a further repeater in the Relay Room.

The trackside signalling cables are prone to damage, and it is possible that one of the conductors could be in contact with the earth if the insulation is damaged. This fault is not self revealing and in isolation does not result in any safety problems. However, if the power supply in the Location case also had an earth fault then a serious problem would occur.



There would be an alternative path between the two earth connections shorting out the Track Relay contact. The TPR would then be permanently energised, indicating track clear, even if it was occupied - a wrongside failure. The technique to overcome this problem is known as double cutting, an additional contact of the controlling relay is added to the negative connection to the controlled relay.



Any trackside circuits must be double cut. In order for a wrongside failure of this type to happen, there now have to be four separate earth faults together, an acceptable risk. Circuits totally within a Location Case, or totally within a Relay Room do not usually need to be double cut, they may be only *single cut*.

TYPES OF TRACK CIRCUIT

There are several types of track circuit available, some of which are described in this document.

The d.c. track circuit

Probably the most widely found track circuit is the d.c. version. The power source in the old days would have been a battery, but in modern installations would be a transformer/rectifier set. The relay would be of d.c. design with a coil to suit the track circuit requirements. Modern designs would use a BR 930 design of relay, but with a specially designed lower voltage coil instead of the standard 50 V version.

The a.c. track circuit

The basic a.c. track circuit operates at the UK normal frequency of 50 Hz, with a transformer providing the power at the feed end. However, the relay works on a different principle from d.c. relays. It incorporates a metal vane and operates in a manner similar to that of a domestic electricity meter. The vane rotates in one direction to close the front contacts, and in the other direction to open them. In order to provide sufficient power, the relay has two coils. One is permanently energised (termed the *local coil*), and the other is connected to the rails (the *control* coil). The feed and the local coil must both be energised from the same a.c. source, but the current in the two coils must be 90° out of phase for the relay to operate efficiently. This phase difference is achieved by feeding the track circuit through a capacitor at the feed end.

It is possible to design a relay with only a single operating coil, but this would result in much larger currents flowing through the rails which would lead in turn to greater problems with leakage current across the ballast.

Track circuits using audio frequencies are also available, using special techniques to safely generate and detect the frequencies required.

BONDING

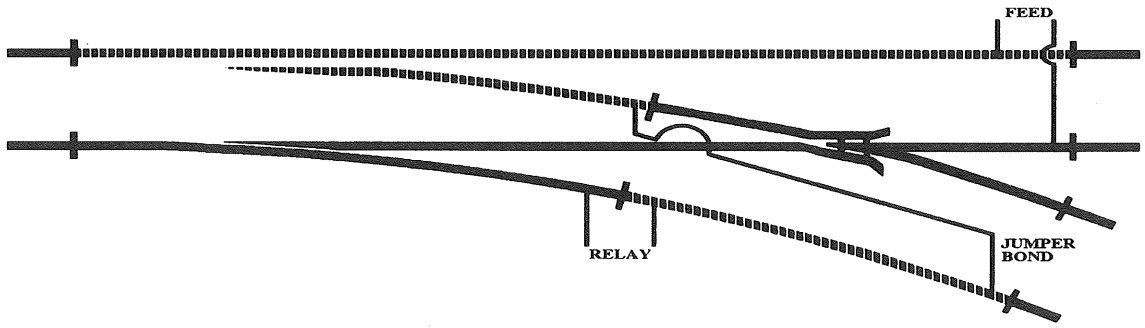
Bonding is a general term used to describe the practice of electrically connecting various rails together for signalling purposes. There are two categories of bonding.

Fishplate Bonding

This is the simplest form of bonding and is applied around each (non-insulated) fishplate along the length of the track circuit. The purpose is simply to provide a reasonable low resistance path round each fishplate. The bond is usually made up of two galvanised steel wires attached to the rails with tapered pins driven into holes in the rail web.

Jumper Bonding

Jumper bonding is a more complex subject, concerned closely with tracks in point and crossing areas. Considering a simple point - the physical arrangement of the rails results in a dead short circuit across the track feed wherever it is put. It is therefore necessary to install extra insulated joints in the rails, not for marking the extremities of the track circuit, but just to enable it to work.



To make things easier to understand in the diagram, the rails connected to one side of the power supply are drawn dotted, whilst those connected to the other side are drawn 'solid'. A wheel anywhere within the limits of the track circuit must always connect a 'dotted' rail and a 'solid' rail together. A jumper bond is necessary to join the two separated 'dotted' rails together.

Bonding must try and ensure that the current flowing from the feed to the relay and then back to the feed passes through as much of the rails as possible, although there usually has to be a compromise. The diagram shows one way of doing it, and in this example, the current flows through every part of the 'dotted' rail, but not every part of the 'solid' rail. In addition, if a bond falls off, the relay must de-energise.

The two additional fishplates required could just as easily have been placed in the 'straight' rails rather than in the 'curved' rails in the diagram above. In practice they would normally be placed in the lesser used of the two diverging lines to reduce maintenance needs.

TRACTION SYSTEMS

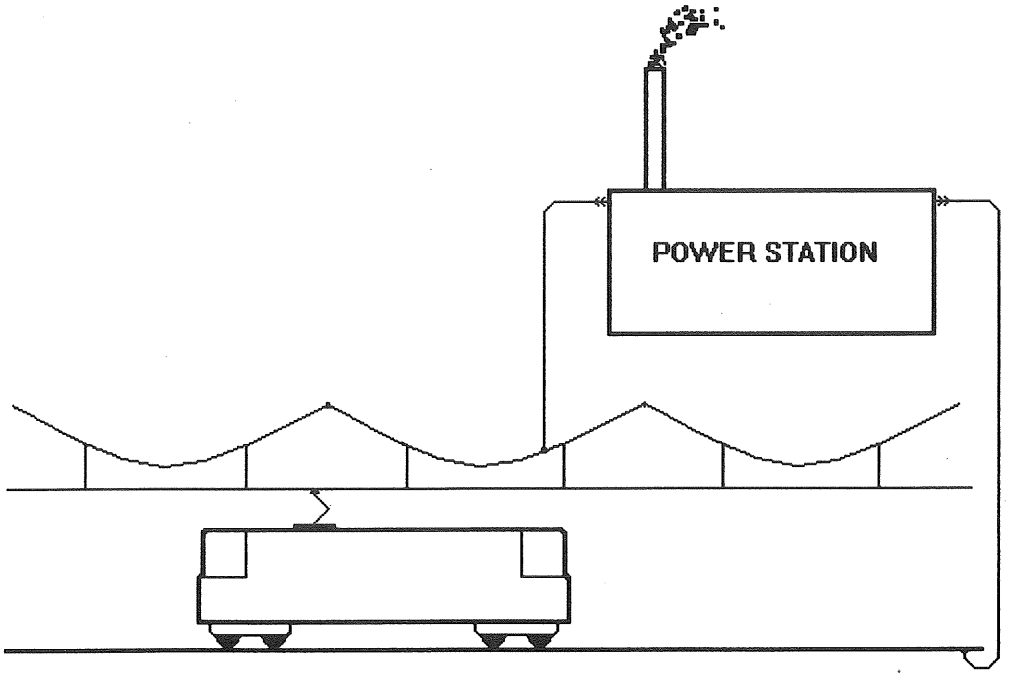
There are two electric traction systems used in the UK. One using d.c., was introduced in the 1920s and is found covering a large area of England south and west of London. There is also a large system in the Liverpool suburban area. The London Underground uses a basically similar system with a small, but significant, variation as far as signalling is concerned.

All main line electrification, except where the existing d.c. system is being extended, uses an a.c. system, and this will be discussed first.

The a.c. System

The a.c. system uses a voltage of 25,000 volts, or 25 kV, the frequency being the frequency of the national electrical system - in the UK 50 Hz. This system was developed by the French in the 1950s and has now become virtually a world standard. The current is picked up by the motive power units from an overhead wire, commonly but erroneously known as the catenary. The Overhead Line Equipment (*OLE*) - the official term - consists of a horizontal copper *contact wire* from which the current is collected. This is suspended from the *catenary*, the curved wires suspended from masts or gantries at regular intervals. 'Catenary' is actually a mathematical term describing the curve that a very fine chain forms when suspended from two spaced points.

The current is collected from the contact wire by a *pantograph* fitted to the vehicle. This is a sprung frame supporting a pan incorporating a carbon strip which rubs along the contact wire:



Although horizontal, the contact wire actually zigzags either side of the centre line of the track. This is done deliberately to even out the wear on the carbon strip on the pantograph, and to prevent a groove being worn in it. Arms on each mast pull and push the contact wire from one side to the other.

Electrical power is measured in Watts by the multiple of the voltage and current. Thus for a given power required from the vehicle, the higher the voltage, the lower the current and vice versa. In this system the very high voltage results in small currents, and this means that the contact wire and all feeder cables can be of small diameter. This in turn saves considerably in cable costs - copper being an expensive commodity. Another advantage of the high voltage/low current system is that the electrical substations supplying the overhead line equipment may be spaced many miles apart.

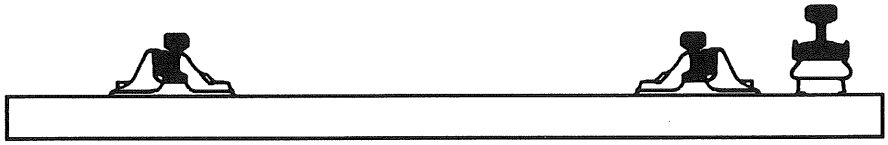
Having picked up the current from the contact wire it then passes to the control equipment and motors. However, the current must complete the path ultimately back to the power station. The only way to achieve this would be to either have a second contact wire which is quite impractical (although this can be seen on one or two 3-phase traction systems of great antiquity) or to use the rails. Thus having passed through the traction equipment the current flows through the wheels and into the rails to reach the power station. As the track is in good contact with the earth through the sleepers and ballast a proportion of the return current actually passes through the earth as well.

However the main problem is that the rails now have both the traction return current and the track circuit current flowing in them together. The insulated joints which isolate the track circuits will obviously not help the traction currents very much! These problems will be discussed below after a description of the d.c. traction system.

One other item concerning a.c. traction system causes problems with the signalling, but not directly related to track circuits. If a conductor is running parallel to another conductor and one of the conductors is carrying a.c., then a.c. will be generated in the second conductor. This effect is known as *induction* and the current is said to be 'induced' into the second conductor. With a.c. traction systems exactly this situation exists. The contact wire carries a.c. at a high voltage, and parallel to this are the trackside signalling cables at ground level. Thus a.c. is induced into all the signalling cables. These currents can give rise to two problems - one involving signalling safety and one involving Health and Safety. The signalling safety is ensured by using a.c. *immune* relays - that is relays that cannot be energised by a.c., only d.c. The Health and Safety problem is that quite high voltages can build up on long circuits, which could be dangerous to technicians working on the system. The longer the circuit the high the induced voltage will be. This problem is resolved by limiting the length of all circuits to 2 km which keeps the induced voltages down to sensible levels. Any circuits which need to be longer than 2 km have to be broken by inserting a repeater relay.

The d.c. System

On Railtrack the d.c. traction system does not use an overhead line, but a *third rail* mounted on the side of the track, outside and above one of the running rails.



Shoes on the train hang down and rub along the top of the rail to pick up the current. Problems appear with bad contact in days of frost, the wrong sort of snow and the much-feared Autumn leaves. On the Docklands light railway the pick-ups on the vehicles actually contact the underside of the conductor rail which eliminates most of the problems.

Although initially the d.c. system is cheap and easy to install - just laying in the third rail - it does have severe disadvantages. Because of the low voltage (650 V - 700 V) the currents are very high, in the order of several thousand amps. This in turn requires extremely fat feeder cables and frequent substations along the trackside.

Once again the traction return currents have to return through the rails. London Underground eliminate the whole problem by laying a fourth rail in the centre of the track for the traction return. On London Underground, therefore, the running rails are reserved solely for the track circuits and life becomes easy.

Many countries have d.c. systems using overhead lines. The most common voltages found on main lines are 1,500 or 3,000 V, whilst lower voltages are frequently used on isolated suburban tramway type lines. D.C. metro systems in the UK where street running is involved of course cannot use a third rail, so again an overhead line system is necessary.

THE TRACTION PROBLEM

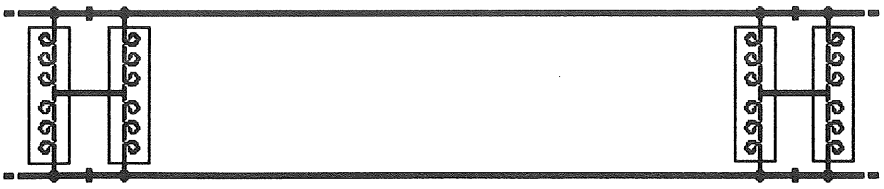
The combination of traction and track circuit currents in the rails leads to two problems:

- The traction currents, even under fault conditions, must not affect the workings of the track circuit. If the traction currents were to keep a track relay energised when the track was occupied that would be a very serious wrongside failure.
- The insulated joints required for the track circuits prevent the return of the traction currents.

The first problem is solved by the fact that if a.c. and d.c. are present together in a circuit they behave in an independent manner, not affecting each other. Thus , if the traction system is a.c., the track circuits would be d.c. and vice versa. The d.c. track relay is specially designed so that it operates only on d.c. and will not be energised by the a.c. traction supply. Such a relay is said to be *a.c. immune*. Similarly, with d.c. traction, a.c. track circuits may be used, again in this case the relays not being operated by the d.c. traction supply.

In some areas a.c. and d.c. traction systems are both present in the same place. The main lines out of Euston station towards Watford Junction are electrified on the a.c. system, whilst the suburban lines which run parallel are on the d.c. system. Although for most of the line there is no physical connection between them, the fact that a proportion of the return traction currents pass through the earth means that there is still a danger of interference between the two systems. Again electrical theory comes to the rescue - a.c. of different frequencies present together in a circuit in general behave in an independent manner. The only exception to this is where one frequency is a multiple of the other. Thus an a.c. frequency of 100 Hz would be affected by an a.c. frequency of 50 Hz and vice versa, whereas frequencies of 51 Hz and 100 Hz would not effect each other. Track circuits in areas where both traction systems are present can thus be a.c. (and so be unaffected by d.c. traction) of such a frequency that will not be affected by the 50 Hz traction system. Such a track circuit is said to be *dual immune*.

The second problem may be dealt with in different ways. In d.c. traction areas, consequently with a.c. track circuits, an electrical device named an impedance is used. An impedance can be thought of as a special type of resistance that only affects a.c., it having negligible effect on d.c. What is more it can be made particularly effective over a range of a.c. frequencies. Thus an impedance can be designed so that it offers quite high resistance to a.c. but virtually no resistance to d.c. flow. Such an impedance consists of a large copper coil immersed in oil and is known as an *impedance bond.*

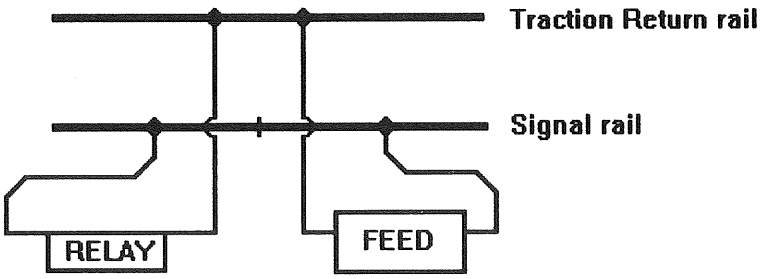


An impedance bond is connected across the rails at each end of every track circuit. (In the diagram above the track feeds and track relays are omitted to simplify the diagram.) A connection is taken from the centre of each bond to a similar point on the bond on the adjacent track circuit, the other side of the insulated joint. The bonds offer a high resistance to the 50 Hz track circuits and effectively isolate one from the other. The d.c. traction current,

however, can flow through the bonds to and from their centre points without hindrance. The traction return currents can thus pass around the insulated joints whilst the track circuit currents are blocked. The impedance bonds are housed in cast iron boxes mounted on the track between the rails.

It is also possible to use impedance bonds to parallel rails in multi-track areas to provide a better return path for the large traction currents, again without interfering with the safe operation of the track circuits.

In a.c. traction areas, with d.c. track circuits the return traction currents are much smaller and it is possible to use one rail for the traction return, without joints, and one rail with joints (the '*signal rail*').



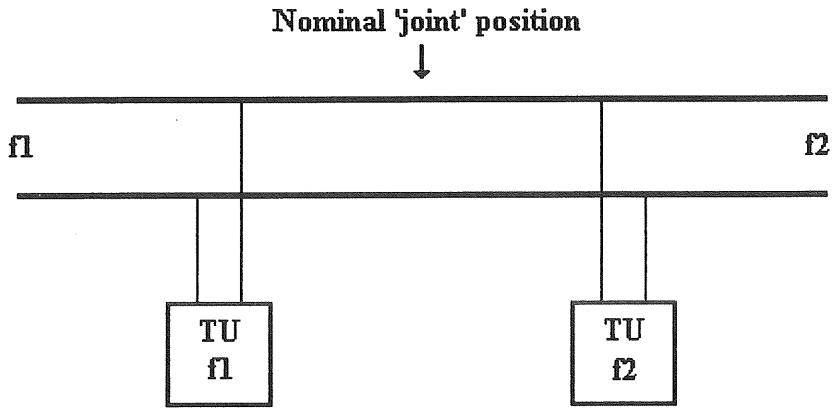
The diagram above shows the 'joint' between two track circuits. The upper rail has no joint and carries the traction current, whilst the lower rail has a joint for track circuit purposes. Notwithstanding the fact that there is no joint in the upper rail the track circuits on either side behave just as if there was one fitted.

JOINTLESS TRACK CIRCUITS

Track circuits are separated from each other by insulated joints. However the railway Civil Engineers do not like ordinary rail joints, let alone the insulated variety. All joints need a lot of expensive maintenance and have to be regularly inspected. The Civil Engineer prefers to weld the rails together in long lengths and eliminate all the joints.

If insulated joints are required in welded rail, the preferred method is to use a glued joint, rather than the conventional bolted type. Two short lengths of rail are glued together under controlled factory conditions. The joint is then taken out and welded in place, resulting in a very reliable joint needing no maintenance.

Another solution is to use track circuits that do not need physical insulated joints at all. *Jointless track circuits* use a.c. of a particular frequency chosen from a set of perhaps eight. To ensure no interference between adjacent track circuits the frequencies are chosen in rotation so that tracks of a particular frequency are always separated by tracks of other frequencies. The tracks are separated from each other by 'electric joints' making use of *tuning units* containing capacitors and inductors. These components, together with the electrical properties of the rails themselves, produce impedancies which both offer high resistance to one particular frequency, and low resistance to a second frequency.



The diagram shows two such tuning units forming a 'joint' separating two track circuits, one of frequency f1 and one of frequency f2. Considering the left hand track circuit, the right hand tuning unit is designed to act as a short circuit at frequency f1. The combination of the left hand tuning unit and the two lengths of rail between the tuning units (connected together by the short circuit produced by the right hand unit) produces a high impedance to block frequency f1. Similarly, when the right hand track circuit, at f2, is considered, the two tuning units and the rails between them act to block frequency f2.

These electric joints do not give a finite demarcation between one track circuit and the next like an insulated joint. There is a length of a few metres overlap at each 'joint' where it is not clear which track is actually occupied, both track relays being de-energised. On plain track this does not pose any problem. Jointless track circuits are not normally installed in point and crossing areas.

There is only one type of jointless track circuit approved for use on a.c. electrified lines, the TI21. The frequencies are carefully chosen to be immune from the effects of the a.c. traction system. In fact each track circuit of this type uses two frequencies, regularly switching alternately between them.

TRACK CIRCUIT PROBLEMS

The working of the track circuit depends on good continuous contact between the wheel and rail. Over the last few years, due to changes in technology in other disciplines, a number of problems have emerged giving cause for concern.

The construction of rolling stock has changed considerably, aluminium extrusions being extensively used. This results in much lighter units which tend to bounce along and lose continuous contact with the rails.

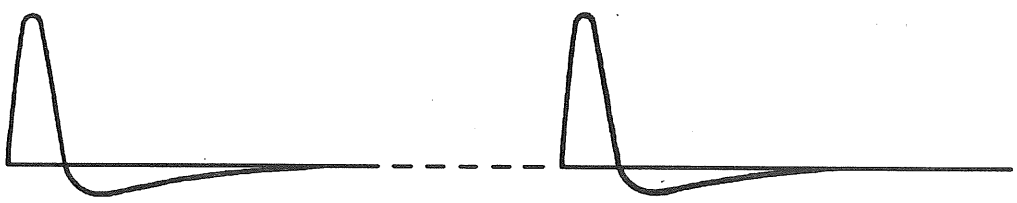
Studies of the wheel/rail interface have improved the smooth running of stock. Previously the stock tended to 'shimmy' along the track moving from side to side as it progressed and this tended to keep the whole of the rail surface clean. Today the stock tends to keep on a much straighter path and the rail surface is not kept as clean.

The change to disc brakes, rather than the older clasp brakes acting directly on the wheel treads, results in the wheel surfaces not being kept so clean.

It has been found that an oxide film tends to build up on the surface of the rail that acts like a Zener diode, that is, its resistance breaks down at a certain voltage. A trainborne *Track Circuit Assistor* (also known as *Track Circuit Actuator*) has been developed to breakdown this resistive film. A single loop of cable mounted under the train acts as the primary winding of a transformer. The secondary winding is formed by the wheel sets and the length of rail between them. The trainborne coil is fed with a.c. at 165 kHz which in turn induces a current of similar frequency in the wheel set/rail loop. The resulting voltage is sufficient to break down the Zener voltage barrier, resulting in near perfect conduction between the wheel and the rail.

Another problem results from changes in traction technology. The latest techniques involve using a.c. induction motors of very simple construction. Their speed of rotation is solely dependant on the frequency of the current applied. Static equipment on the motive power units firstly rectifies the 25 kV supply to d.c. and then invertors convert it back to a.c. at a frequency dependant on the speed required. As a result a large amount of noxious frequencies are generated on the vehicle which pass to the track and interfere badly with the a.c. track circuits, possibly artificially energising the track relay under the train.

There is one particular track circuit that goes a long way to solving some of these problems. Traditionally known as the Jeumont-Schneider track circuit it is now known as the *High Voltage Impulse* track, or HVI for short. The track feed unit applies a series of electrical impulses to the rails at a rate of about 3 every second. Each impulse lasts about one hundredth of a second and is a very distinct shape:



The voltage builds up very rapidly to something in the order of 100 V. It then reduces back to zero equally rapidly, overshoots slightly to a negative voltage and then climbs back to zero.

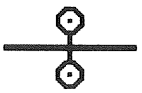
The track relay is specially designed so that it will only remain energised if it receives impulses of the correct shape and, moreover, receives them on a regular basis. The high voltage, even though it is for a short duration overcomes some of the problems with dirty rails and bad contact. The HVI track circuit is not affected by the frequencies generated by traction units as the chance that pulses of the exact shape and regularity would be generated is practically zero.

It is very common practice to use TI21 track circuits on plain track, in combination with HVI track circuits in points and crossing areas.

AXLE COUNTERS

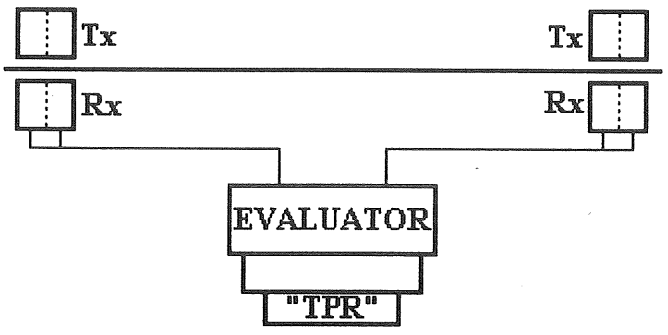
An alternative to the track circuit is the axle counter. The principle of the axle counter is that the total number of axles (although, in fact wheels are detected) entering a section of track are counted. As each axle leaves the section it is again counted and subtracted from the total. If the final result is zero, then the section of track must be clear. Axle counters have the advantage that they are not affected by bad rail/wheel contact, leaves on the line or traction currents in the rails. In addition, they do not need insulated joints in the rails or bonding. However, there is the disadvantage that broken rails are not detected. Axle counters are used in the UK at present to overcome particular problems. They are used on the Forth Bridge where track circuits are adversely affected by the metal deck, and in the very wet tunnels under the River Severn and forming the Mersey Loop. They are now being introduced for general train detection purposes.

The passing of each wheel along one of the rails is detected magnetically by *detector heads*. The symbol used on signalling plans for a detector is



Each detector head consists of a transmitter (Tx) and a receiver (Rx). A Tx on one side of the rail produces a magnetic field around the rail head. An Rx on the other side of the rail can detect this field, which results in an a.c. output from that Rx. As a each wheel approaches and passes the detector head this magnetic field is distorted sufficiently to reverse the output polarity of the Rx. In practice, one detection point consists of two heads spaced a few centimetres apart, and different frequencies are used for the two. This doubling allows the direction of the movement to be determined, which in turn determines whether the axles are counted 'in' or 'out'.

The outputs of the detectors at each end of the section are fed to an *evaluator unit*, which detects and counts the frequency reversals from both ends. A relay, equivalent to a Track Relay, is de-energised unless the wheel count is zero.



If there are points within the section then there will be more than two detectors connected to the evaluator. In these circumstances, looping on the evaluator defines the layout of the detectors to ensure correct 'adding' and 'subtracting'. A detector will usually be involved with two evaluators, one each for the sections on each side, counting out on one and counting in on the other. If the two evaluators are in the same room they can be connected together. If they are not, separate cables from the detector will be required, one to each evaluator.